PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions

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Abstract

Understanding which individual characteristics amplify susceptibility to adversarial opinion manipulation is a core question in cognitive security research, yet real-world data combining structured profiles, diverse attack vectors, and repeated opinion measurements is difficult to obtain and ethically constrained. We address this gap using an ontology-driven multi-agent simulation pipeline in which 100 synthetic pseudoprofiles are crossed with four attack vectors spanning distinct persuasion mechanisms—cognitive reframing, emotional appeal, false consensus, and authority-heuristic exploitation—and four political-opinion leaves in a full factorial design ($n = 1{,}600$ scenarios, Run 8). The primary outcome, adversarial effectivity, measures signed opinion movement in the direction of an adversarially defined goal; mean adversarial effectivity was -14.49 ($SD = 44.40$), with only 32.6% of scenarios producing movement in the intended direction, consistent with pervasive resistance and backfire. Intraclass correlation for absolute shift was ICC(1) = 0.052, indicating that attack–opinion context explains an order of magnitude more variance than stable profile characteristics. In the multivariate profile-moderator model corrected for false discovery, Conscientiousness was the only significant predictor ($b = -3.26$, $SE = 1.14$, $q_{\mathrm{FDR}} = .030$), with higher self-discipline associated with reduced susceptibility across opinion leaves. A leave-one-group-out hierarchical decomposition assigned 23.1% of relative importance to Conscientiousness alone. These results validate a factorial simulation architecture for studying heterogeneity in susceptibility to adversarial persuasion and provide methodological infrastructure for comparison with human experimental data.

1 Introduction

Political influence operations increasingly exploit the gap between information volume and cognitive processing capacity. Rather than relying on technical intrusion, they operate through selective framing, fabricated consensus, pseudo-expert endorsement, and identity-congruent narrative design—mechanisms that shape the conditions of political judgement rather than the content of specific beliefs (Bennett & Livingston, 2018; Lazer et al., 2018; Lewandowsky et al., 2017; Pennycook & Rand, 2019, 2021; Vosoughi et al., 2018). The core question this raises is not simply whether content is false, but whether adversarial exposure predictably shifts opinions in directions that serve strategic goals, and whether that susceptibility varies systematically across individuals.

This question has been increasingly framed in security policy discussions under the term cognitive warfare, recognizing that human attention, trust, and decision-making have become operationally relevant terrain in contemporary conflict and competition (Johns Hopkins University & Imperial College London, 2021; Hung & Hung, 2022; Paulauskas, 2024). We adopt

cognitive sovereignty—the capacity for self-directed political belief formation in the face of adversarial manipulation—as the substantive construct motivating the study design. Susceptibility is defined as the degree to which post-exposure opinions move in directions intended by a hypothetical adversarial operator, rather than as simple opinion change magnitude.

Recent work on large language models (LLMs) as generative agents demonstrates that LLM-driven personas can reproduce human-like social behavior across extended interaction sequences (Park et al., 2023) and that prompted LLM samples can approximate distributional properties of human survey responses (Argyle et al., 2023). These findings support using LLM-generated synthetic profiles as tractable proxies for exploring individual-difference interactions with adversarial exposure, provided results are interpreted as properties of the simulation rather than direct claims about human populations.

A critical design question is why simulation rather than direct profile-to-score elicitation. Prompting an LLM to rate a profile's vulnerability produces a scalar impression that compresses all information into a single opaque judgement. The simulation pipeline produces something fundamentally different: a conditional response surface linking each profile to structured outcomes across attack types, opinion domains, and scoring rules. This yields three specific methodological advantages. (1) Susceptibility is estimated as a conditional, task-dependent property—the same profile may be exploitable for one issue framing or attack mechanism but not another—whereas a one-shot rating compresses all of that into a scalar assuming susceptibility is a single latent trait with no task structure. (2) Repeated indicators across tasks constitute a measurement system: internal consistency, residual covariance, ICC, and SEM paths can be computed and audited, whereas a single elicited score provides no psychometric decomposition. (3) The staged pipeline externalizes the assumptions that determine how profile features map to outcomes, making each link inspectable and comparable to human experimental data independently, rather than remaining hidden inside a one-step inference. The sharpest way to state the distinction: a direct LLM judgement estimates who *looks* vulnerable; a simulation pipeline estimates *under which conditions* a profile becomes directionally exploitable.

The present pilot extends the single-attack design of earlier runs to a full factorial layout, introducing adversarial effectivity as a directional primary outcome and a battery of modern statistical methods including intraclass correlation for variance partitioning, Benjamini–Hochberg FDR correction, and a hierarchical leave-one-group-out R^2 decomposition.

2 Materials and Methods

2.1 Ontology-Constrained Design

Run 8 uses three hierarchical ontologies—PROFILE, ATTACK, and OPINION—each stored as a typed tree and estimated at the leaf level. PROFILE combines continuous leaves (age; Big Five personality facets as percentile scores) with categorical leaves (sex). OPINION samples four leaf nodes from the domain `Defense_and_National_Security`: Alliance Commitment Support, Civil Defense Preparedness Support, Domestic Security Powers Expansion Support, and Strategic Autonomy Support.

ATTACK contributes four leaf nodes spanning distinct persuasion mechanism families:

1. `Social_Media_Misinformation > Misleading_Narrative_Framing` — selective reframing exploiting cognitive heuristics;

2. `Manipulative_Persuasion > Fear_Appeal_Scapegoating_Post` — emotionally charged content activating threat-based processing;
3. `Coordinated_Amplification > Astroturf_Comment_Wave` — synthetic grassroots consensus exploiting social-proof;
4. `Manipulative_Persuasion > Pseudo_Expert_Authority_Cue` — fabricated expert endorsement exploiting authority heuristics.

The design is a complete factorial: each of 100 pseudoprofiles is crossed with all 4 attack leaves and all 4 opinion leaves, producing $100 \times 4 \times 4 = 1{,}600$ attacked scenarios with no no-attack control condition.

2.2 Baseline, Exposure, and Post-Exposure Assessment

For each profile i, attack a, and opinion leaf k, the pipeline elicits a baseline opinion and a post-exposure opinion on a signed integer scale from $-1{,}000$ to $+1{,}000$. Exposure texts are generated as platform-native manipulative messages tailored to the specific attack–opinion combination, then audited for realism. Post-exposure responses are audited for coherence and boundedness. Every analyzed row has passed a structured review layer designed to detect implausible reversals, coarse rounding, and issue-incoherent response shifts.

2.3 Adversarial Effectivity Outcome

The OPINION ontology encodes per-leaf direction annotations ($d_k \in \{-1, +1\}$) based on an adversarially configured goal for the modeled opinion domain. The signed shift for profile i under attack a on leaf k is

$$\Delta_{iak} = \text{post}_{iak} - \text{baseline}_{iak}, \tag{1}$$

and the primary outcome, adversarial effectivity, is

$$\text{AE}_{iak} = \Delta_{iak} \times d_k. \tag{2}$$

Positive AE means the attack moved the opinion in the adversarially intended direction; negative values indicate backfire.

2.4 Profile-Level Susceptibility Index

The repeated-outcome SEM estimates profile-level path coefficients from a four-indicator model in which each indicator is the adversarial effectivity for one opinion leaf averaged across attack vectors:

$$\overline{\text{AE}}_{i \cdot k} = \alpha_k + X_i \beta_k + C_i \gamma_k + \varepsilon_{ik}. \tag{3}$$

The post hoc susceptibility index aggregates target-conditional ridge task models across all 16 attack–opinion combinations:

$$S_i^*(\mathcal{T}) = \text{PctRank}\left(\sum_{t \in \mathcal{T}} w_t \hat{A}_{it} \right), \quad w_t \propto \frac{n_t}{\text{CV-MSE}_t}. \tag{4}$$

2.5 Statistical Analysis Plan

ICC(1) (Shrout & Fleiss, 1979) partitions variance into between-profile and within-profile components. A repeated-outcome SEM simultaneously regresses four attack-averaged opinion indicators on profile moderators with freely estimated residual covariances. Moderator p-values are adjusted using the Benjamini–Hochberg FDR procedure (Benjamini & Hochberg, 1995). A leave-one-group-out R^2 decomposition quantifies the marginal contribution of each feature group to the susceptibility index.

3 Results

3.1 Descriptive Overview

Run 8 generated 1,600 attacked scenarios (100 profiles × 4 attacks × 4 opinion leaves). Mean absolute shift was 43.16 ($SD = 17.86$), mean signed shift was 29.48 ($SD = 36.23$), and mean adversarial effectivity was −14.49 ($SD = 44.40$). Only 32.6% of scenarios produced positive adversarial effectivity. Mean attack realism was 0.71 and mean post-exposure plausibility was 0.74. Figure 1 summarizes the factorial structure.

3.2 Variance Partitioning

ICC(1) for absolute shift was 0.052 ($\sigma^2_{\text{between}} = 16.62$, $\sigma^2_{\text{within}} = 302.81$); for adversarial effectivity, ≈ 0.000. Fewer than 6% of outcome variance is between-profile: attack–opinion context explains the remainder.

3.3 Factorial Outcome Surface and Moderation Strength

Figure 2 visualizes inter-individual variability (standard deviation of adversarial effectivity across 100 profiles) across the complete 4 × 4 attack–opinion matrix. Cells with high SD reveal where individual-difference moderation has the greatest reach; cells with low SD indicate uniform response regardless of profile. Attack vectors differed: Misleading Narrative Framing was most effective (mean AE = −11.98, 35.8% positive), followed by Fear Appeal Scapegoating (−13.14, 35.5%), Astroturf Comment Wave (−15.80, 31.0%), and Pseudo Expert Authority Cue (−17.05, 28.2%). Strategic Autonomy Support was the only opinion leaf with positive mean adversarial effectivity (+29.98); Civil Defense Preparedness (−36.64) and Domestic Security Powers Expansion (−39.31) showed strong resistance. The absolute-shift distribution by opinion leaf is shown in Figure 3; the attack-vector comparison panel in Supplementary Figure S4.

3.4 Structural Equation Model

The repeated-outcome SEM (Figure 5) converged with CFI = 1.000, RMSEA = 0.000 ($n = 100$, four opinion indicators averaged across attack vectors). Strongest leaf-specific paths: Conscientiousness → Alliance Commitment ($b = -6.76$, $p = .006$); Conscientiousness → Civil Defense Preparedness ($b = -4.54$, $p = .013$); Baseline Score → Domestic Security ($b = -6.53$, $p < .001$); Exposure Quality → Strategic Autonomy ($b = 4.80$, $p = .010$). Full path coefficients are in Supplementary Table S5.

3.5 Profile Moderator Analysis

In the multivariate model (Table 2; Supplementary Table S2), Conscientiousness was the sole predictor surviving FDR correction ($b = -3.26$, $SE = 1.14$, $p = .004$, $q_{\text{FDR}} = .030$). Univariate: Conscientiousness ($b = -2.51$, $p = .009$, $q = .065$); Openness trend ($b = 1.39$, $p = .096$, n.s. after FDR).

3.6 Hierarchical Importance Decomposition

Full-model CV-$R^2 \approx 0$ confirming weak aggregate profile prediction at pilot scale. Relative importance: Big Five 32.0%, Demographics 28.8%, Conscientiousness alone 23.1%. Supplementary Figure S5 shows the hierarchical breakdown; Figure 4 shows individual moderator weights.

Figure 1. Full factorial study design for Run 8 ($100 \times 4 \times 4 = 1{,}600$ scenarios).

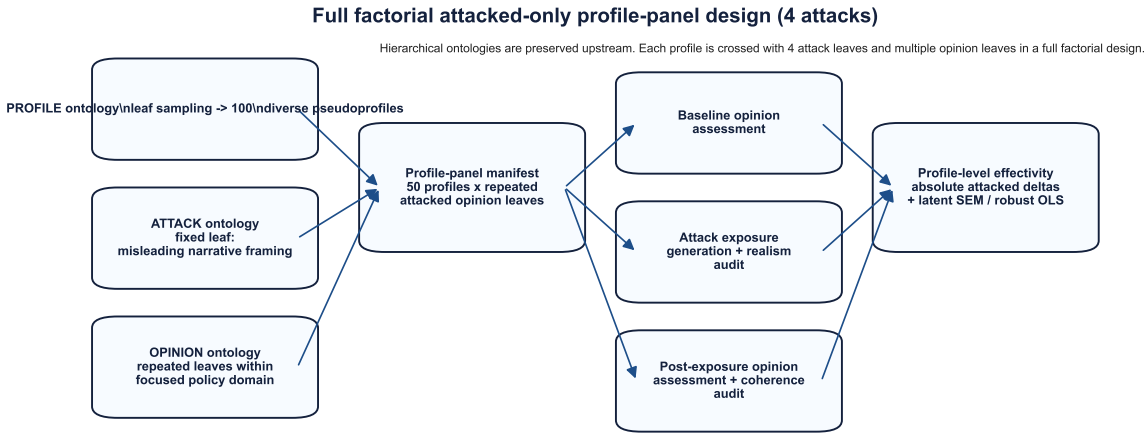


Table 1. Attacked effectivity descriptive statistics by repeated opinion leaf.

opinion_leaf_label	n_rows	mean_baseline	mean_post	mean_signed_delta	mean_abs_delta	sd_abs_delta	mean_adv_eff	sd_adv_eff
Civil_Defense_Preparedness_Support	400	351.842	388.480	36.638	45.443	18.056	-36.638	32.384
Strategic_Autonomy_Support	400	416.065	446.045	29.980	45.035	17.664	29.980	37.965
Alliance_Commitment_Support	400	447.688	459.685	11.998	41.562	14.241	-11.998	42.265
Domestic_Security_Powers_Expansion_Support	400	192.702	232.012	39.310	40.595	20.435	-39.310	22.810

Note. Descriptive statistics aggregated across 4 attack vectors per opinion leaf. Adversarial effectivity weights shift by the per-leaf adversarial goal direction.

Table 2. Profile moderators of attacked effectivity: controlled contrasts, SEM paths, and descriptive susceptibility weights.

Moderator	Role	Controlled mean b	Controlled p	Ridge mean b	Weight %	SEM mean b	SEM mean |b|	SEM min p	SEM paths
Big Five Conscientiousness Mean %	core	-2.511	0.009			-2.470	3.179	0.006	4.000
Big Five Neuroticism Mean %	core	-0.026	0.979			0.473	0.682	0.374	4.000
Sex Other	core	0.622	0.777	1.081	5.221	1.246	4.942	0.075	4.000
Sex Female	core	0.413	0.833	-0.696	2.809	1.400	4.708	0.095	4.000
Big Five Openness To Experience Mean %	core	1.395	0.096			0.899	1.478	0.179	4.000
Big Five Extraversion Mean %	exploratory	-0.986	0.340						
Big Five Agreeableness Mean %	exploratory	-0.277	0.735						

Note. Controlled coefficients summarize moderator associations with mean attacked effectivity. SEM columns summarize the repeated-outcome path model across the attacked opinion leaves. Weight percentages come from the target-conditional regularized aggregation used to compute the post hoc susceptibility index.

Figure 2. Inter-individual moderation strength across the 4×4 attack–opinion factorial. Height = standard deviation of adversarial effectivity across 100 profiles. Higher bars indicate cells where individual differences drive the largest spread in manipulation outcomes.

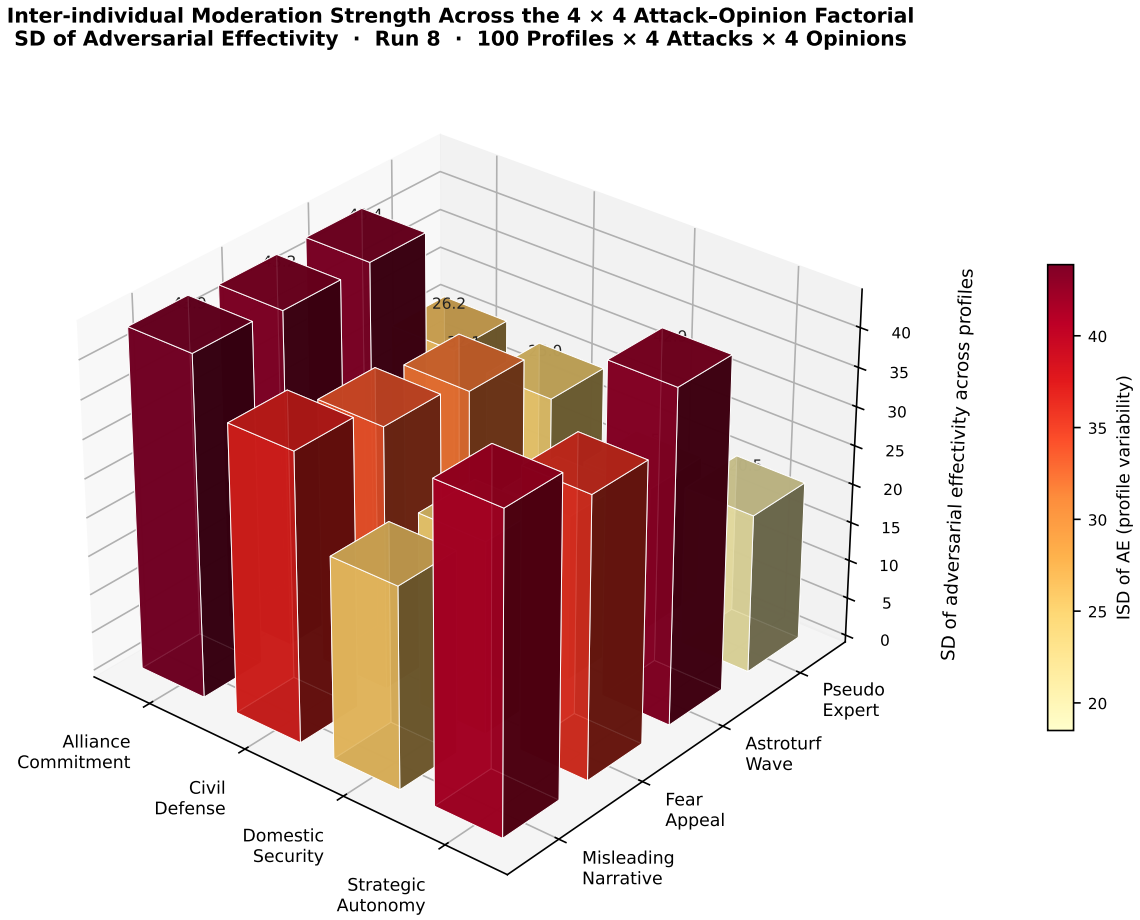


Figure 3. Absolute attacked opinion shift by opinion leaf. Violins show density; bars show interquartile range; diamonds mark means; points show individual profile–attack averages ($n = 400$ per leaf).

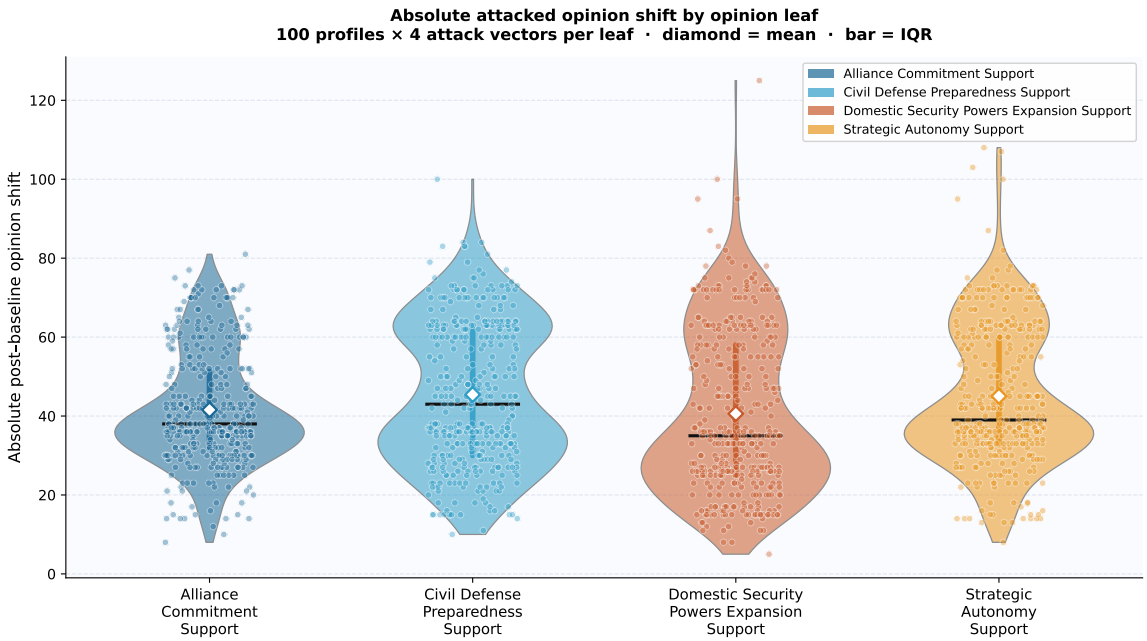


Figure 4. Normalized susceptibility weights across profile moderators (Run 8).

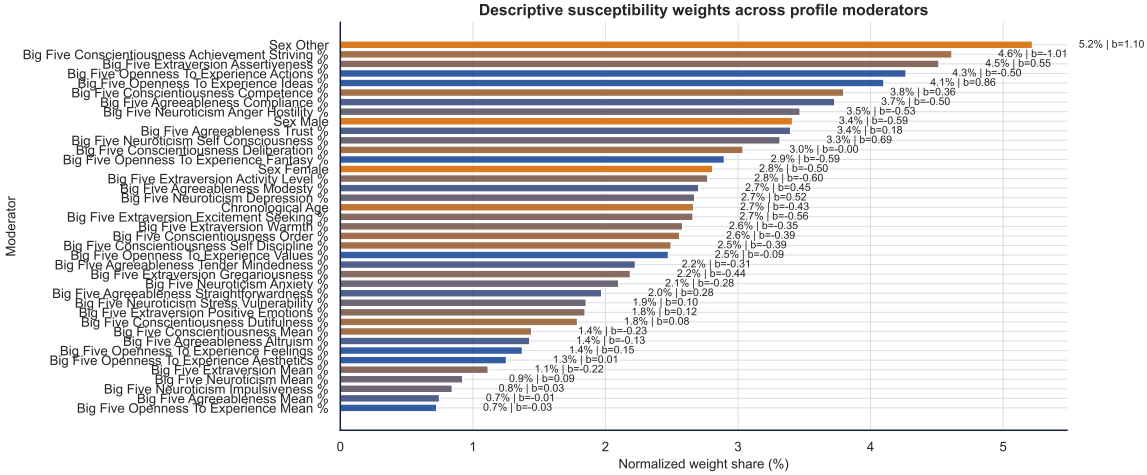
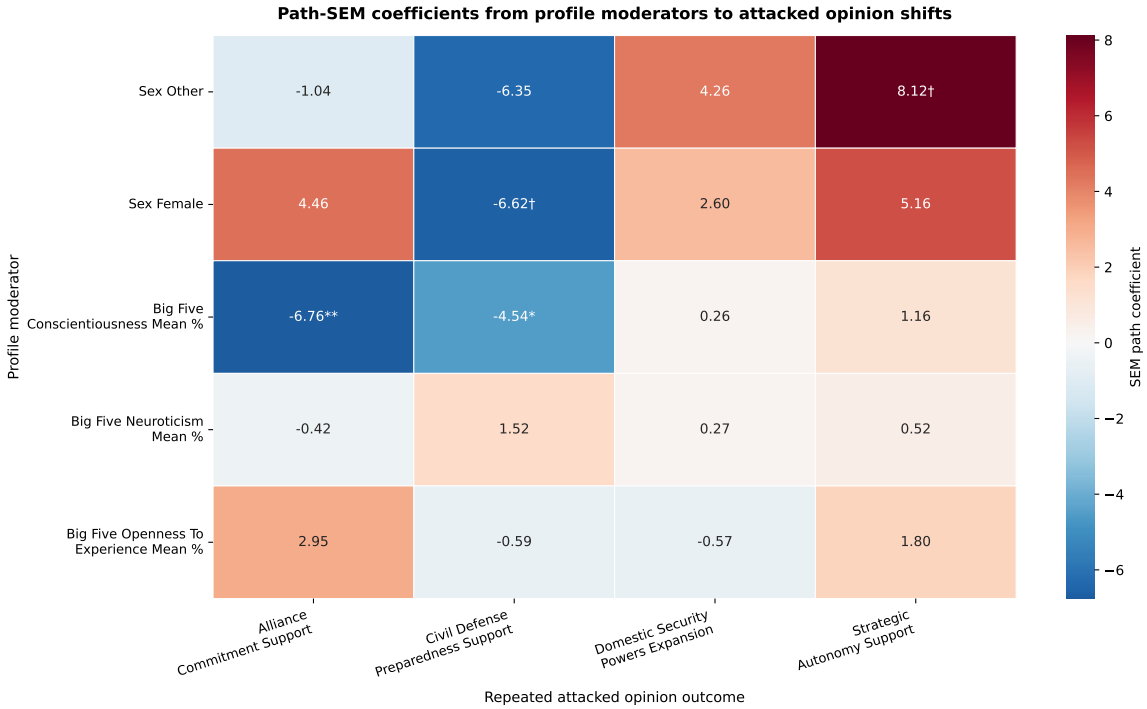


Figure 5. Repeated-outcome path-SEM coefficients across attacked opinion indicators. Cells show path coefficients from profile moderators to attack-averaged opinion-shift indicators. Stars mark $p < .05$; daggers mark $p < .10$.



4 Discussion

4.1 Attack–Opinion Context Dominates Profile-Level Variance

Near-zero ICC values—0.052 for absolute shift, ≈ 0 for adversarial effectivity—establish that attack–opinion context is the primary determinant of outcome variance. This finding aligns with evidence that even psychologically targeted content produces modest individual-level effects (Matz et al., 2017), and it constrains the practical utility of profile-based susceptibility classification: the same individual may resist one attack on one issue while being substantially moved by a different combination.

4.2 Conscientiousness as the Dominant Moderator

Conscientiousness emerged as the most consistent predictor of reduced adversarial effectivity across the multivariate model, univariate comparison, SEM leaf-specific paths, and the hierarchical importance decomposition—the only moderator surviving FDR correction, accounting for nearly a quarter of relative importance. The negative sign is interpretable: deliberation, goal-directed self-regulation, and preference for carefully vetted information plausibly confer a dispositional buffer against rapid adversarial opinion displacement, consistent with evidence that analytic cognitive style reduces susceptibility to partisan misinformation (Pennycook & Rand, 2019, 2021).

4.3 Simulation as Conditional Measurement, Not Scalar Judgement

A critical consideration for interpreting these results is what the simulation pipeline produces relative to the alternatives. Direct one-shot elicitation—prompting an LLM to rate how susceptible a profile is—produces a single scalar judgement compressing all person, attack, and issue information into an opaque response. The simulation pipeline produces a structured measurement system: 16 task-specific conditional susceptibility estimates, repeated opinion indicators suitable for SEM, and an auditable staging of baseline, exposure, and post-exposure states. This object can be compared to human experimental data in ways that a scalar cannot: through criterion validity against observed opinion shifts, convergent validity with psychometric scales, and stability testing under prompt and seed variation. The object produced is not merely "more data"—it is the right *functional form* of the construct. Susceptibility is not a scalar person-property; it is a conditional function $S_i(a, o, c)$ where the same profile may be exploitable under one attack–issue combination and resilient under another. Simulation preserves that structure; one-shot judgement collapses it.

4.4 Directional Asymmetry and Opinion-Specific Patterns

Only Strategic Autonomy Support showed positive mean adversarial effectivity, while Civil Defense and Domestic Security opinions moved consistently against the intended adversarial direction. This directional asymmetry is consistent with psychological reactance under threat-focused framing (Roozenbeek et al., 2022), though whether this pattern reflects genuine psychological dynamics or LLM training-distribution artifacts cannot be determined without human criterion data.

4.5 Limitations

The profiles are synthetic and full-model CV-R^2 is near zero, confirming weak aggregate profile prediction at pilot scale. Standard errors do not account for the clustered structure (16 observations per profile); future work should apply cluster-robust inference (Cameron et al., 2008). Near-perfect SEM fit is expected in a near-saturated four-indicator, 100-observation model. Criterion validity against observed human opinion shifts remains the necessary next step.

5 Conclusion

This pilot demonstrates that a full factorial ontology-driven simulation provides a methodologically coherent architecture for studying susceptibility to adversarial persuasion. Five contributions over earlier single-attack pilots: (1) a four-vector attack taxonomy spanning cognitive, emotional, social-proof, and authority-based mechanisms; (2) adversarial effectivity as a directional outcome distinguishing successful manipulation from backfire; (3) ICC diagnostics to establish the variance-partitioning ceiling for profile-level moderation; (4) Benjamini–Hochberg FDR correction for the moderator battery; (5) hierarchical leave-one-group-out R^2 decomposition for the susceptibility index. The dominant finding—Conscientiousness as the most consistent moderator across multiple analytic lenses—is theoretically coherent with deliberative processing accounts of persuasion resistance. The present pilot provides the methodological infrastructure for comparison with human experimental data; substantive claims about real populations require that validation.

A Supplementary Materials

Figure 6. Baseline versus post-attack opinion scores across all 1,600 scenarios.

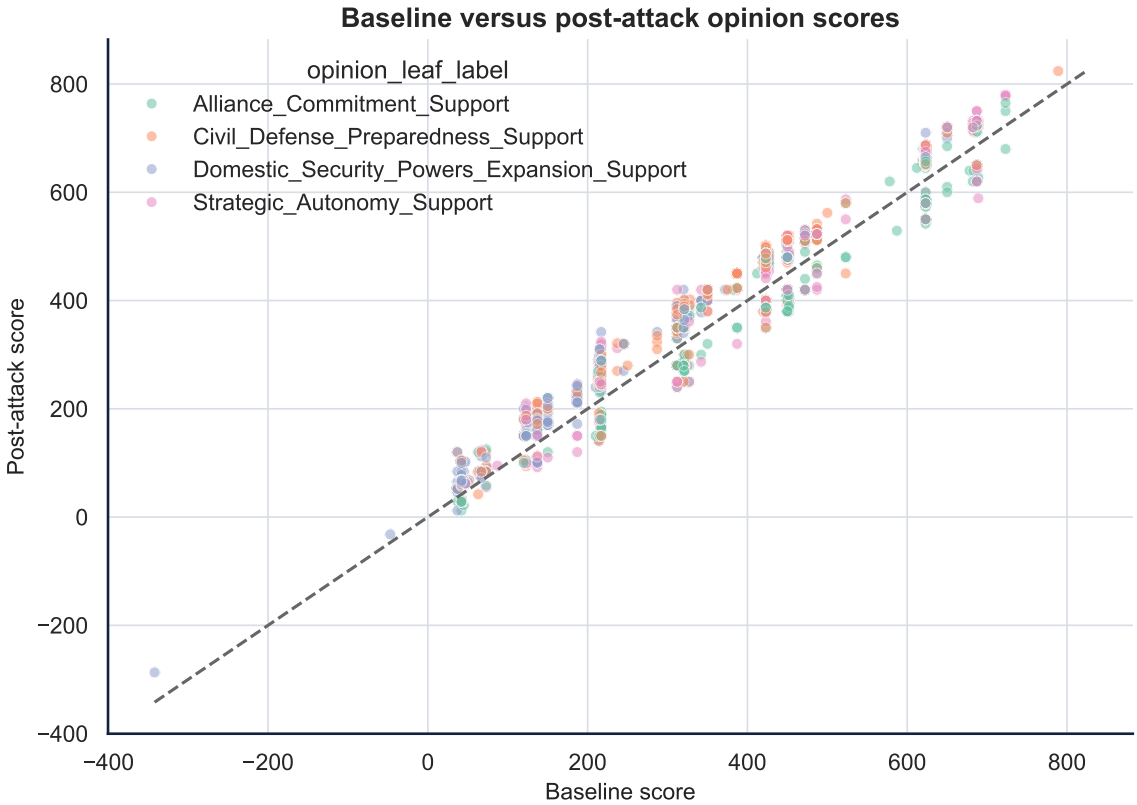


Figure 7. Per-profile attack effectivity heatmap (100 profiles × 4 opinion leaves; ordered by susceptibility index).

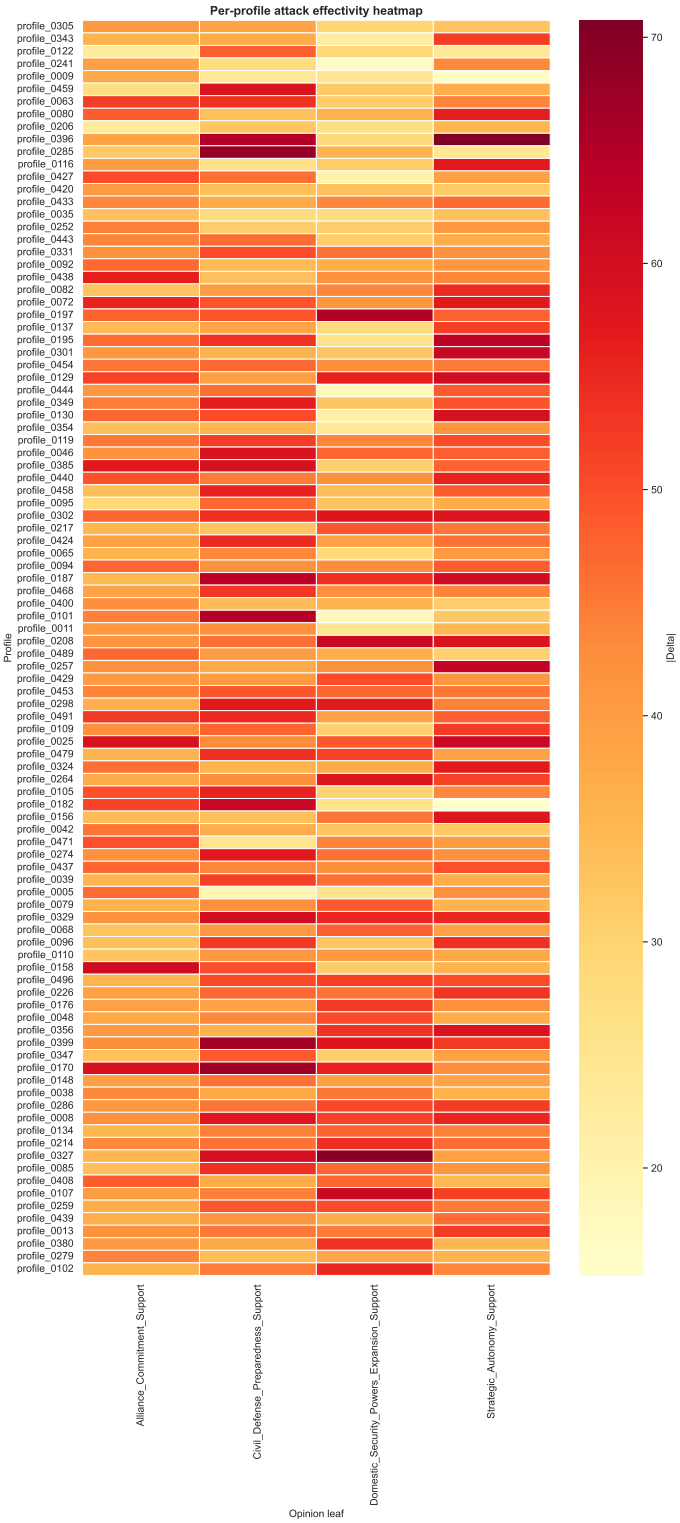


Figure 8. Distribution of the post hoc empirical susceptibility index across 100 profiles.

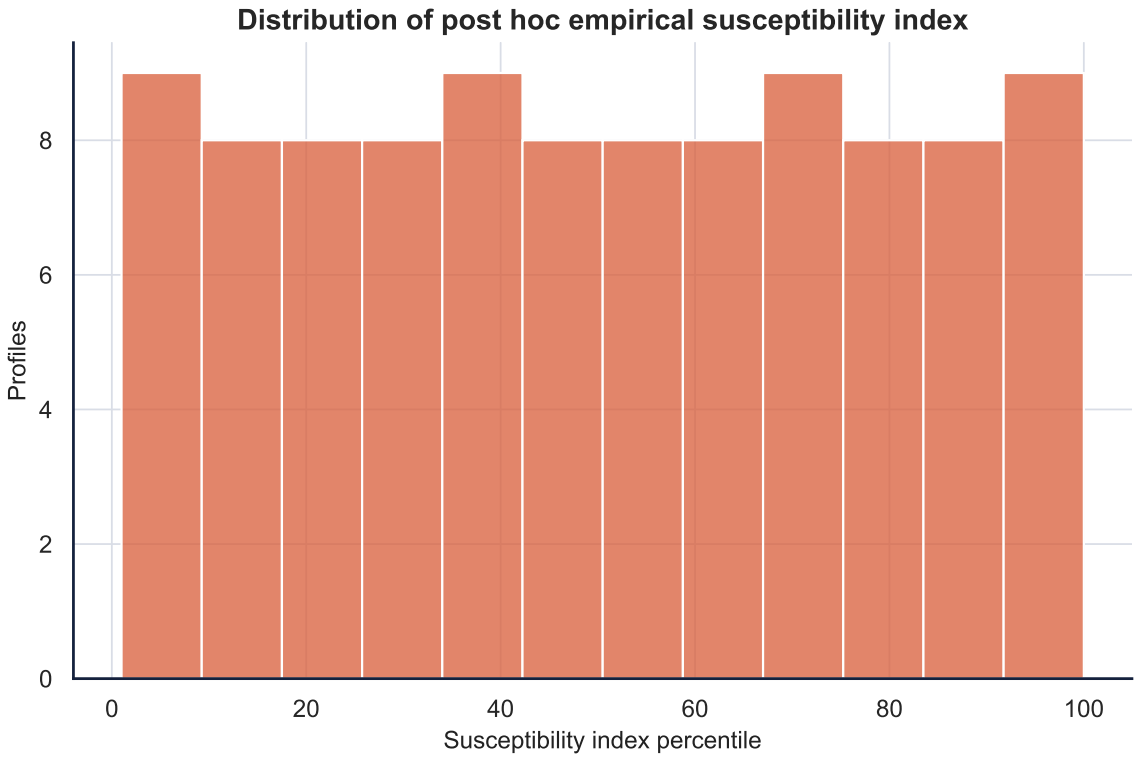


Figure 9. Adversarial effectivity by attack vector.

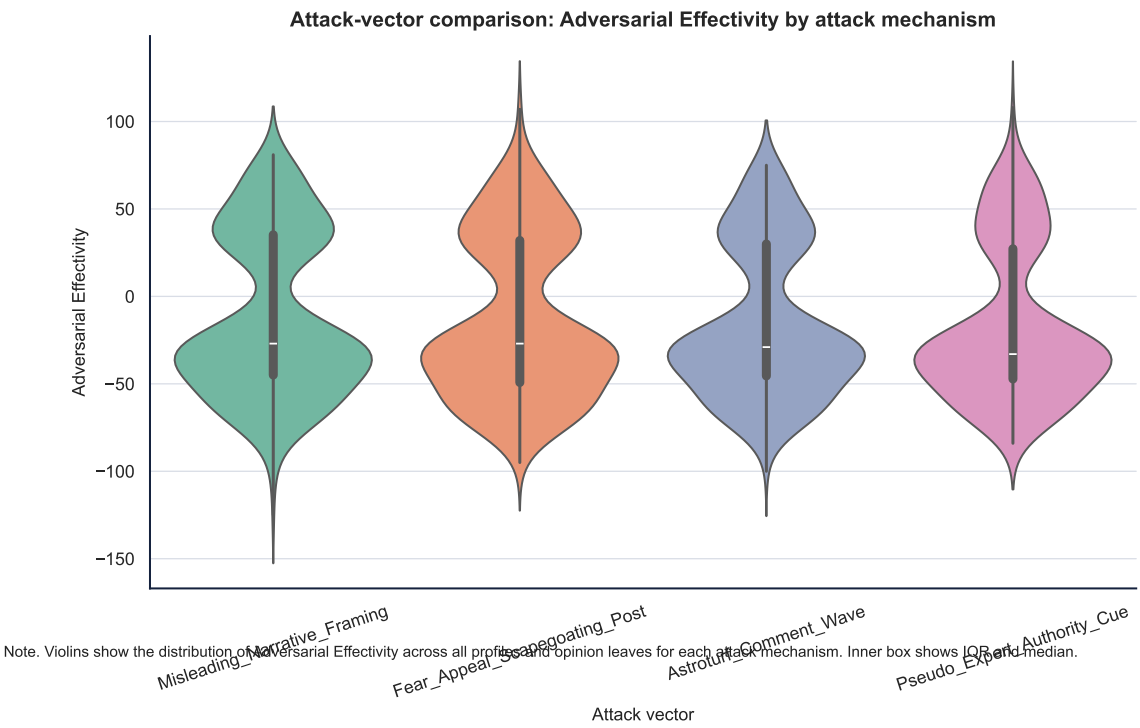
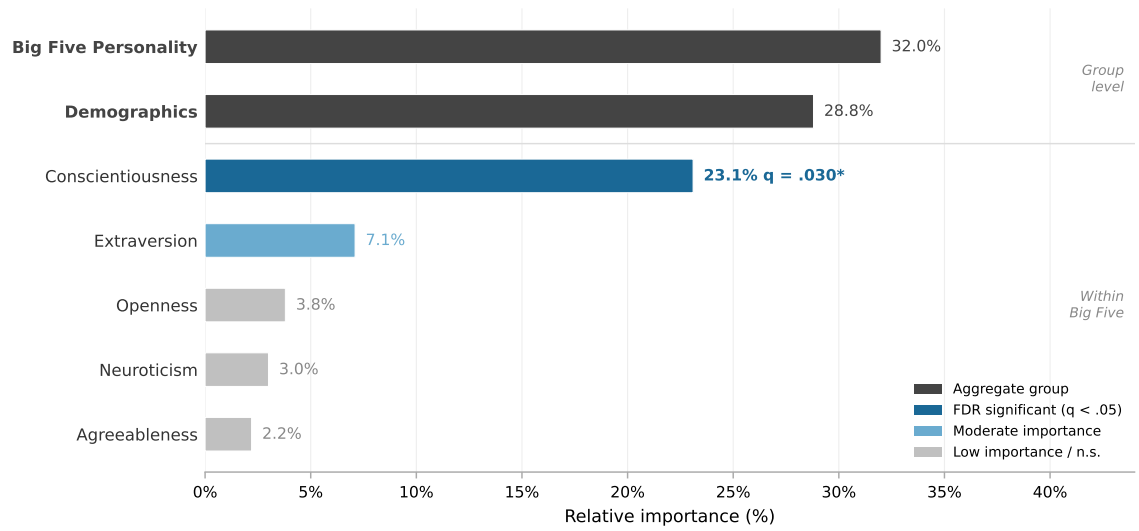


Figure 10. Hierarchical feature importance in the conditional susceptibility model.



** Benjamini–Hochberg FDR corrected q-value from the multivariate profile-moderator model. Relative importance = |marginal CV-R²| /Σ|marginal CV-R²| × 100 (leave-one-group-out decomposition of the conditional susceptibility index).*

Table 3. Ontology leaves used in run_8.

Ontology	Role	Domain / Parent	Leaf
ATTACK	Factorial attack leaf	Social Media Misinformation	Misleading Narrative Framing
ATTACK	Factorial attack leaf	Manipulative Persuasion	Fear Appeal Scapegoating Post
ATTACK	Factorial attack leaf	Coordinated Amplification	Astroturf Comment Wave
ATTACK	Factorial attack leaf	Manipulative Persuasion	Pseudo Expert Authority Cue
OPINION	Repeated indicator	Defense and National Security	Alliance Commitment Support
OPINION	Repeated indicator	Defense and National Security	Civil Defense Preparedness Support
OPINION	Repeated indicator	Defense and National Security	Domestic Security Powers Expansion Support
OPINION	Repeated indicator	Defense and National Security	Strategic Autonomy Support

Note. Full ontology paths are condensed to domain and leaf for readability. The ATTACK ontology contributes 4 factorial leaves; the OPINION ontology contributes 4 repeated indicator leaves.

Table 4. Core and exploratory profile moderator comparison.

Moderator	Role	Multivariate b	Multivariate p	Univariate b	Univariate p	Ontology group	Normalized weight %
Big Five Conscientiousness Mean %	core	-3.258	0.004	-2.511	0.009	Personality: Conscientiousness	1.443
Big Five Neuroticism Mean %	core	1.273	0.261	-0.026	0.979	Personality: Neuroticism	0.923
Sex Other	core	2.395	0.357	0.622	0.777	Demographics: Sex	5.221
Sex Female	core	2.017	0.368	0.413	0.833	Demographics: Sex	2.809
Big Five Openness To Experience Mean %	core	0.567	0.499	1.395	0.096	Personality: Openness	0.727
Big Five Extraversion Mean %	exploratory	-2.353	0.130	-0.986	0.340	Personality: Extraversion	1.115
Big Five Agreeableness Mean %	exploratory	-0.422	0.602	-0.277	0.735	Personality: Agreeableness	0.748

Note. Core terms are entered into the latent SEM or primary multivariate profile model. Normalized weight percentages summarize each moderator's share of the total fitted importance after accounting for the moderator's observed variability.

Table 5. Assumption register and peer-review risk register.

Type	Item	Status	Mitigation
Assumption	Pilot sample size is sufficient for stable moderation estimates.	ok	Treat estimates as exploratory and scale scenario count in later runs.
Assumption	Attack-only design is intentional and aligned with the moderation question.	ok	Interpret results as moderation among attacked individuals, not as a no-attack counterfactual contrast.
Assumption	Opinion measurement has enough numeric resolution.	risk	Prompt enforces high-resolution scores in [-1000,1000].
Assumption	Attack text realism is adequate.	ok	Self-supervised realism reviewer + rewrite loop in Stage 03.
Assumption	SEM converges with acceptable fit.	ok	Use robust OLS supplement, reduce heterogeneity, and increase scenario count before substantive interpretation.
Assumption	LLM outputs are reproducible and auditable.	ok	Persist all prompts/outputs in provenance/raw_llm.
Critique	Synthetic LLM agents may not represent human cognition or causal response behavior.	addressed	Added transparent assumption register and explicit exploratory caveat in reports.
Critique	Adversarial content may be unrealistic or too generic.	addressed	Added realism review agent with rewrite loop and heuristic checks.
Critique	Model dependence on a single LLM could bias outputs.	addressed	Model is CLI-configurable and run metadata captures exact model for replication.
Critique	Insufficient sample size inflates uncertainty and may destabilize SEM.	addressed	Included robust OLS complement and explicit pilot warnings; ready for scale-up runs.
Critique	Attack-only designs cannot estimate the incremental effect of exposure relative to a no-attack counterfactual.	addressed	The report states explicitly that the attacked-only design estimates heterogeneity of post-minus-baseline response among attacked individuals only.

Note. The pilot is methodological and exploratory. Risks are surfaced explicitly so later scale-up runs can target the current design bottlenecks.

Table 6. Reproducibility manifest for run_8.

Field	Value
run_id	run_8
paper_title	PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions
output_root	[project root]/evaluation/run_8
report_root	[project root]/research_report/report
report_assets_root	[project root]/research_report/assets
n_scenarios	10
n_profiles	100
seed	88
attack_ratio	1.000
attack_leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
attack_leaves	Misleading_Narrative_Framing, Fear_Appeal_Scapegoating_Post, Astroturf_Comment_Wave, Pseudo_Expert_Authority_Cue
focus_opinion_domain	Defense_and_National_Security
max_opinion_leaves	4
profile_candidate_multiplier	5
primary_moderator	posthoc_profile_susceptibility_index
bootstrap_samples	800
use_test_ontology	True
ontology_root	[project root]/src/backend/ontology/separate/test
openrouter_model	mistralai/mistral-small-3.2-24b-instruct
temperature	0.150
max_repair_iter	2
profile_generation_mode	deterministic
self_supervise_attack_realism	True
realism_threshold	0.760
self_supervise_opinion_coherence	True
coherence_threshold	0.760
generate_visuals	True
export_static_figures	True
build_report	True
resume_from_stage	01
stop_after_stage	09
save_raw_llm	True
timeout_sec	90
max_concurrency	20
run_stage_checks	False
stage_check_scenarios	3

Note. The manifest captures the full pipeline configuration used to generate the pilot outputs and manuscript assets.

Table 7. Leaf-specific path-SEM coefficients.

Outcome leaf	Moderator	SEM b	SE	p
Adversarial Delta Indicator Alliance Commitment Support	Big Five Neuroticism Mean %	-0.417	2.291	0.856
Adversarial Delta Indicator Alliance Commitment Support	Big Five Openness To Experience Mean %	2.954	2.201	0.179
Adversarial Delta Indicator Alliance Commitment Support	Big Five Conscientiousness Mean %	-6.758	2.448	0.006
Adversarial Delta Indicator Alliance Commitment Support	Sex Female	4.457	5.300	0.400
Adversarial Delta Indicator Alliance Commitment Support	Sex Other	-1.042	5.543	0.851
Adversarial Delta Indicator Alliance Commitment Support	Mean Baseline Abs Score	-1.116	2.831	0.694
Adversarial Delta Indicator Alliance Commitment Support	Mean Exposure Quality Score	-2.318	2.252	0.303
Adversarial Delta Indicator Civil Defense Preparedness Support	Big Five Neuroticism Mean %	1.524	1.713	0.374
Adversarial Delta Indicator Civil Defense Preparedness Support	Big Five Openness To Experience Mean %	-0.589	1.645	0.720
Adversarial Delta Indicator Civil Defense Preparedness Support	Big Five Conscientiousness Mean %	-4.541	1.830	0.013
Adversarial Delta Indicator Civil Defense Preparedness Support	Sex Female	-6.616	3.962	0.095
Adversarial Delta Indicator Civil Defense Preparedness Support	Sex Other	-6.348	4.144	0.126
Adversarial Delta Indicator Civil Defense Preparedness Support	Mean Baseline Abs Score	-1.202	2.117	0.570
Adversarial Delta Indicator Civil Defense Preparedness Support	Mean Exposure Quality Score	-1.285	1.684	0.445
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Big Five Neuroticism Mean %	0.268	1.170	0.819
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Big Five Openness To Experience Mean %	-0.568	1.124	0.613
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Big Five Conscientiousness Mean %	0.256	1.250	0.838
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Sex Female	2.595	2.706	0.338
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Sex Other	4.257	2.830	0.133
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Mean Baseline Abs Score	-6.528	1.445	0.000
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Mean Exposure Quality Score	0.382	1.150	0.740
Adversarial Delta Indicator Strategic Autonomy Support	Big Five Neuroticism Mean %	0.519	1.884	0.783
Adversarial Delta Indicator Strategic Autonomy Support	Big Five Openness To Experience Mean %	1.800	1.809	0.320
Adversarial Delta Indicator Strategic Autonomy Support	Big Five Conscientiousness Mean %	1.161	2.013	0.564
Adversarial Delta Indicator Strategic Autonomy Support	Sex Female	5.162	4.358	0.236
Adversarial Delta Indicator Strategic Autonomy Support	Sex Other	8.119	4.557	0.075
Adversarial Delta Indicator Strategic Autonomy Support	Mean Baseline Abs Score	0.204	2.328	0.930
Adversarial Delta Indicator Strategic Autonomy Support	Mean Exposure Quality Score	4.800	1.852	0.010

Note. Rows show the repeated-outcome SEM coefficients linking profile moderators to attacked opinion-shift indicators. Each opinion indicator is the adversarial delta averaged across the four attack vectors; attack-specific variation is examined in Supplementary Figure S4.

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